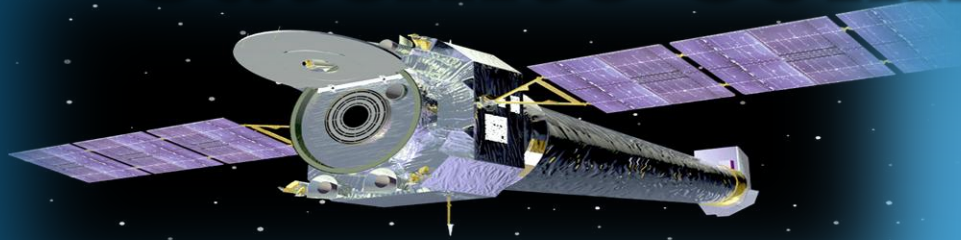


Satellite Communication



MBEYA UNIVERSITY OF SCIENCE AND
COMMUNICATION



Introduction

- Satellite communication has become a part of everyday life. We make international as easily as local calls, see live football matches and other events.
- This capability to exchange information is made possible by satellite.
- It stretches back to an era when the term satellite was merely an idea conceived by a few inspired individuals.
- These pioneers included authors such as Arthur C. Clark, who in 1945 fostered the idea of a worldwide satellite system.

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- In most instances, satellite systems offer more flexibility than submarine cables, buried underground cable, microwave radio or optical fiber system.

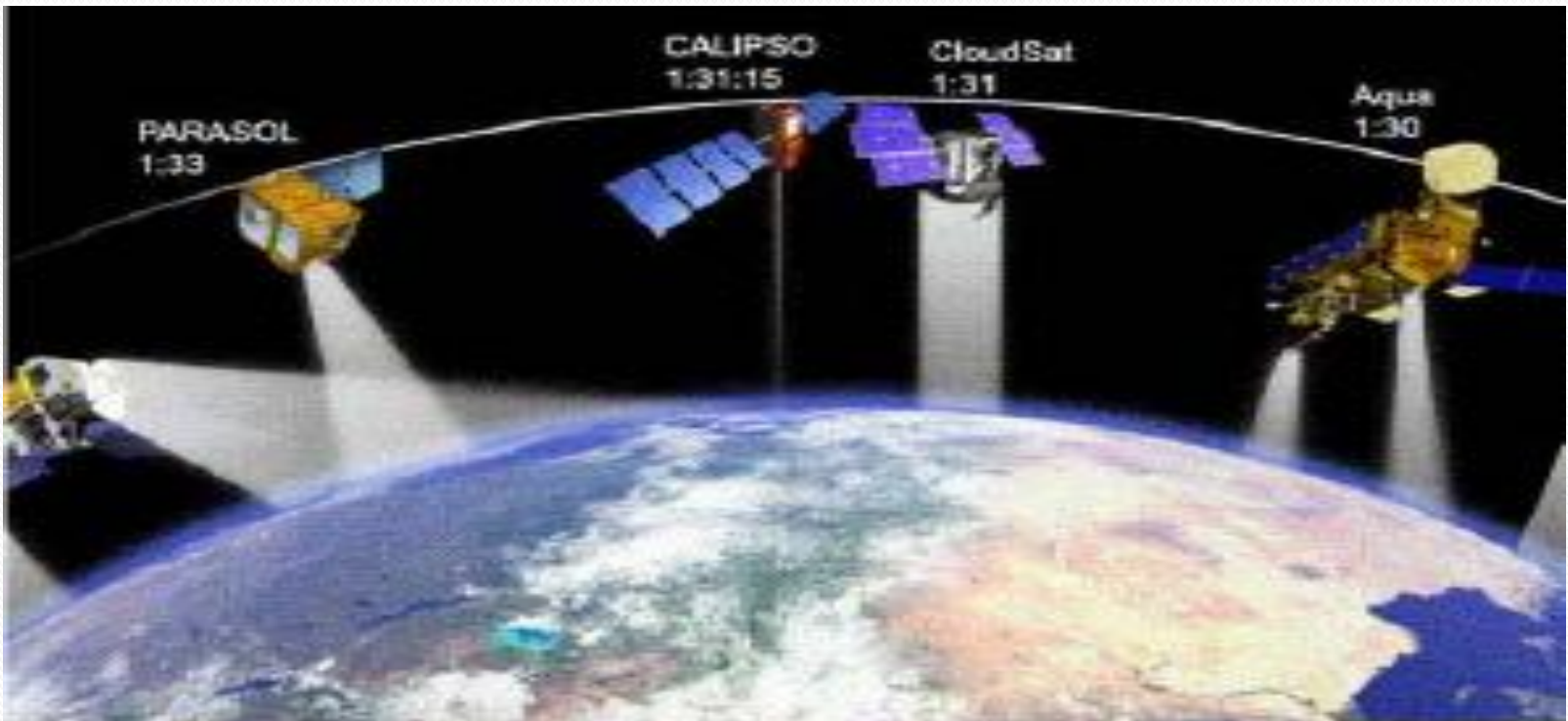
What exactly is a satellite?

- The word satellite originated from the Latin word “**Satellit**”- meaning an attendant, one who is constantly hovering around & attending to a “master” or big man.
- For our own purposes however a satellite is simply any body that moves around another (usually much larger) one in a mathematically predictable path called an orbit.

What is satellite communication?

- A satellite communication is basically an electronic communication package placed in orbit whose prime objective is to initiate or assist another through space.
- A communication satellite is a station in space that is used for telecommunication, radio and television signals.
- The first satellite with radio transmitter was in 1957.

Communication Satellites



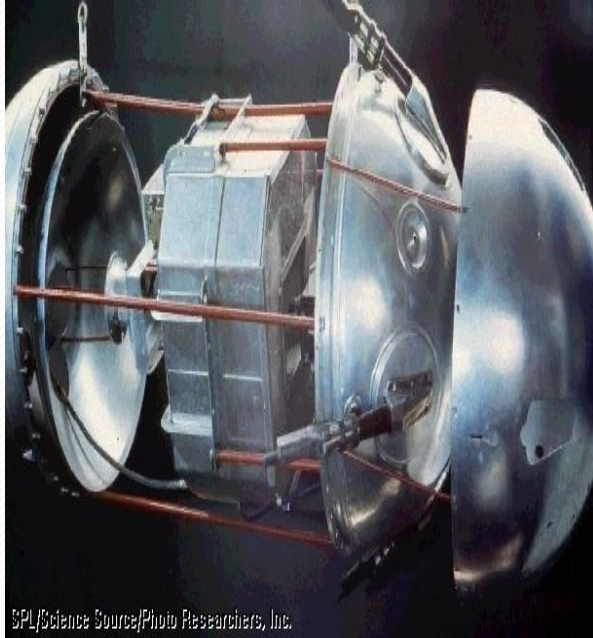
History of satellite

- The concept of using object in space to reflect signals for communication was proved by Naval Research Lab in Washington D.C. when it use the Moon to establish a very low data rate link between Washington and Hawaii in late 1940's.
- Russian started the Space age by successfully launching SPUTNIK the first artificial spacecraft to orbit the earth, which transmitted telemetry information for 21 days in Oct. 1957.

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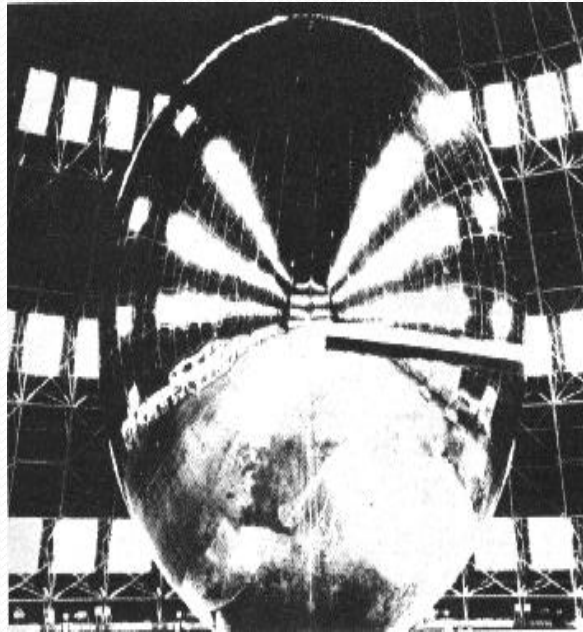
- The American followed by launching an experimental satellite EXPLORER In 1958.
- In 1960 two satellite were deployed “Echo” & “Courier”
- In 1962 AT&T launched *Telstar I*, the first satellite to receive and transmit simultaneously.
- *Telstar II* was successfully launched in 1963. It was used for telephone, television, facsimile and data transmission.
- There are about 750 satellites in the space, most of them are used for communication.

Satellite Pictures

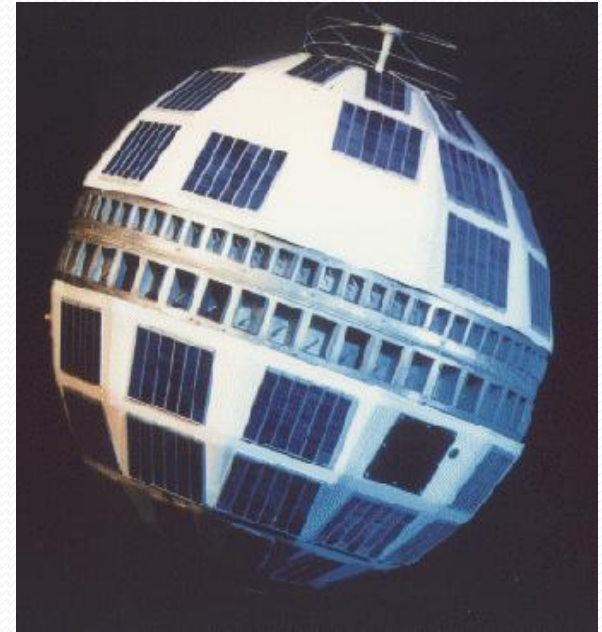


SPL/Science Source/Photo Researchers, Inc.

October 1957, the first artificial satellite **Sputnik - I** was launched by former Soviet Russia.

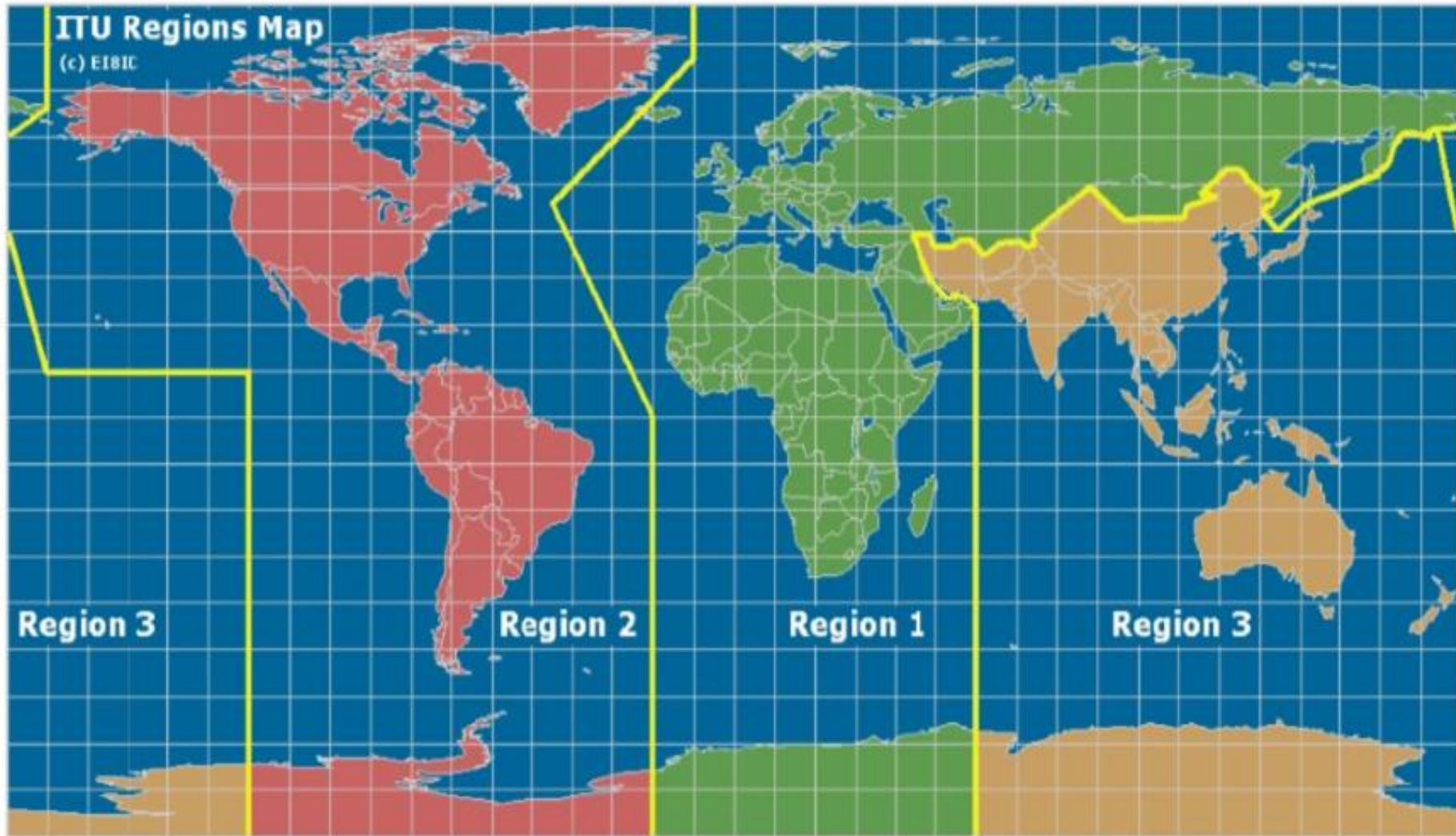


The first artificial passive satellite **Echo-I** of NASA was launched in August 1960.



In July 1962 active satellite **Telstar** was developed and launched

2. ITU radio spectrum regions



Frequency bands

FREQUENCY BANDS

Band	Frequency range	Wavelength range
Extremely low frequency (ELF)	< 3 kHz	>100 km
Very low frequency (VLF)	3 - 30 Hz	10 - 100 krn
Low frequency(LF)	30 - 300 kHz	1 - 10 km
Medium frequency (MF)	300 kHz - 3 MHz	100m - 1km
High frequency (HF)	3 - 30 MHz	10 - 100m
Very high frequency (VHF)	30 - 300 MHz	1 - 10m
Ultra high frequency (UHF)	300 MHz - 3 GHz	10cm - 1m
Super high frequency (SHF)	3 - 30 GHz	1 - 10cm
Extremely high frequency (EHF)	30 - 300 GHz	1mm - 1cm

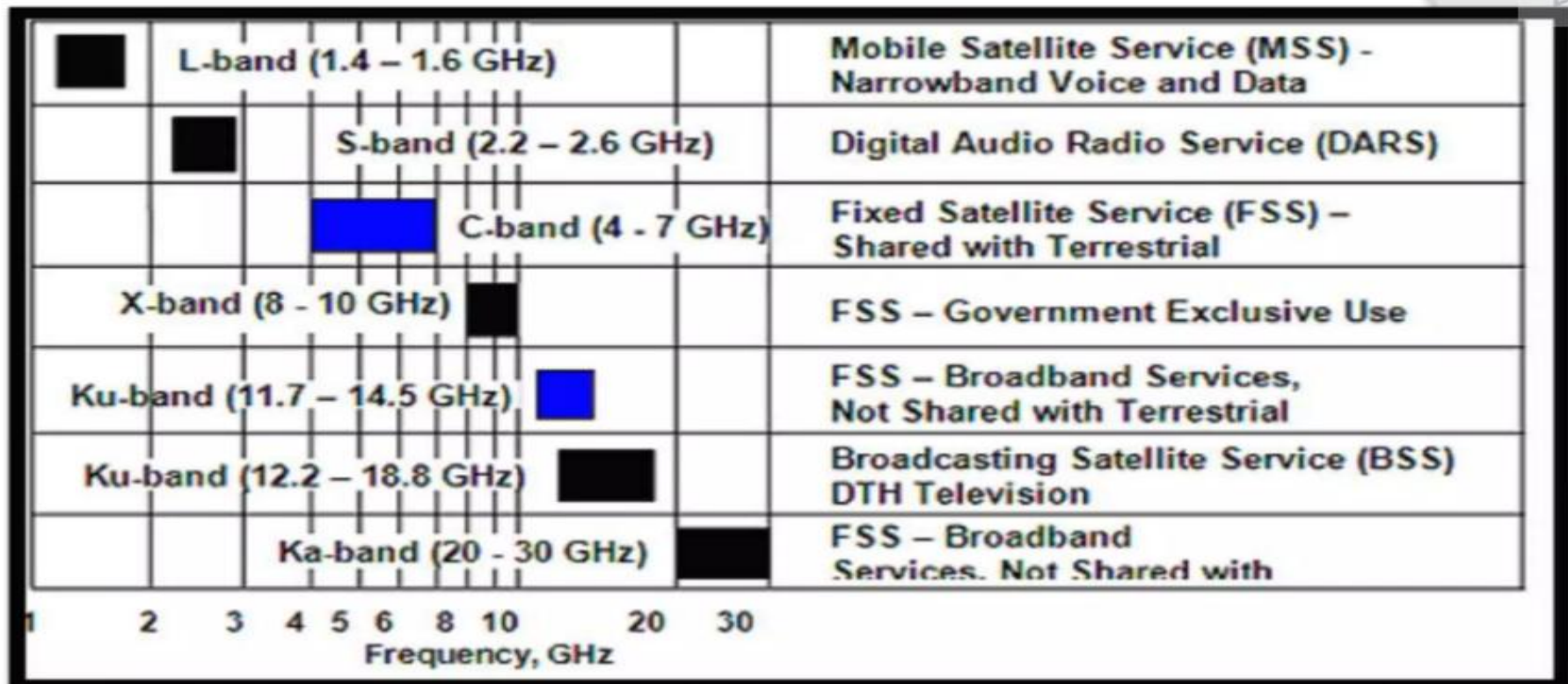
How Satellites are used

- **Service Types**

- Fixed Service Satellites (FSS)
 - Example: Point to Point Communication
- Broadcast Service Satellites (BSS)
 - Example: Satellite Television/Radio
 - Also called Direct Broadcast Service (DBS).
- Mobile Service Satellites (MSS)
 - Example: Satellite Phones

Bands and services

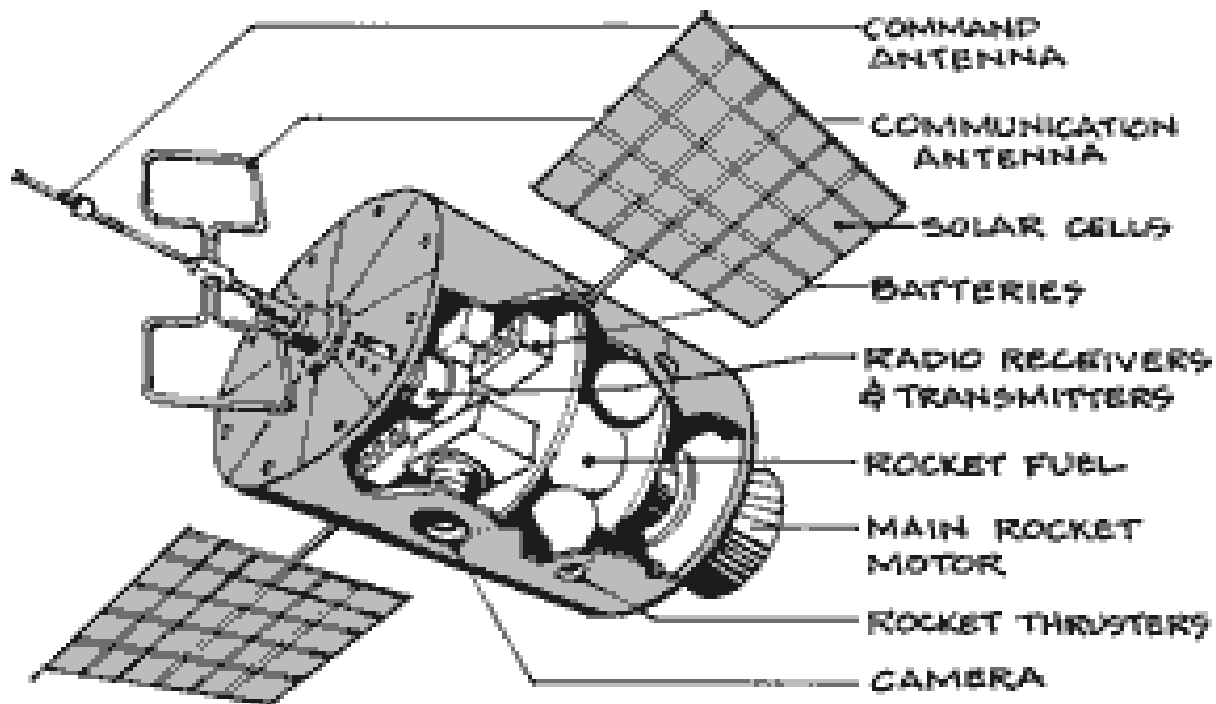
Where used frequency bands:



Application Example

- Telecommunications
- Military communications
- Navigation systems
- Remote sensing and surveillance
- Radio / Television Broadcasting
- Astronomical research
- Weather observation

Components of a satellite



Types of Satellites

- **Passive satellite**
- A passive satellite is one that simply reflects a signal back to earth ; there are no gain devices on board to amplify or repeat the signal. These type of satellites simply reflects or scatters the micro wave signal back to another station.
- An advantage of passive satellites is that they do not require sophisticated electronic equipment on board.
- Example: military and non profitable services.

Satellite Communications Needs

- Space vehicle used as communications platform
(Earth-Space-Earth, Space-Earth, Space-Space)
- Space vehicle used as sensor platform with communications
- Ground station(s) (Tx/Rx)
- Ground receivers (Rx only)

Major problems for satellites

- Positioning in orbit
- Stability
- Power
- Communications
- Harsh environment

Advantages of Satellite

- High channel capacity (>100 Mb/s)
- Low error rates ($P_e \sim 10^{-6}$)
- Stable cost environment (no long-distance cables or national boundaries)
- Wide area coverage (whole North America, for instance)
- Coverage can be shaped by antenna patterns

Disadvantages of Satellite

- Expensive to launch
- Expensive ground stations required
- Cannot be maintained
- Limited frequency spectrum
- Limited orbital space (geosynchronous)
- Constant ground monitoring required for positioning and operational control

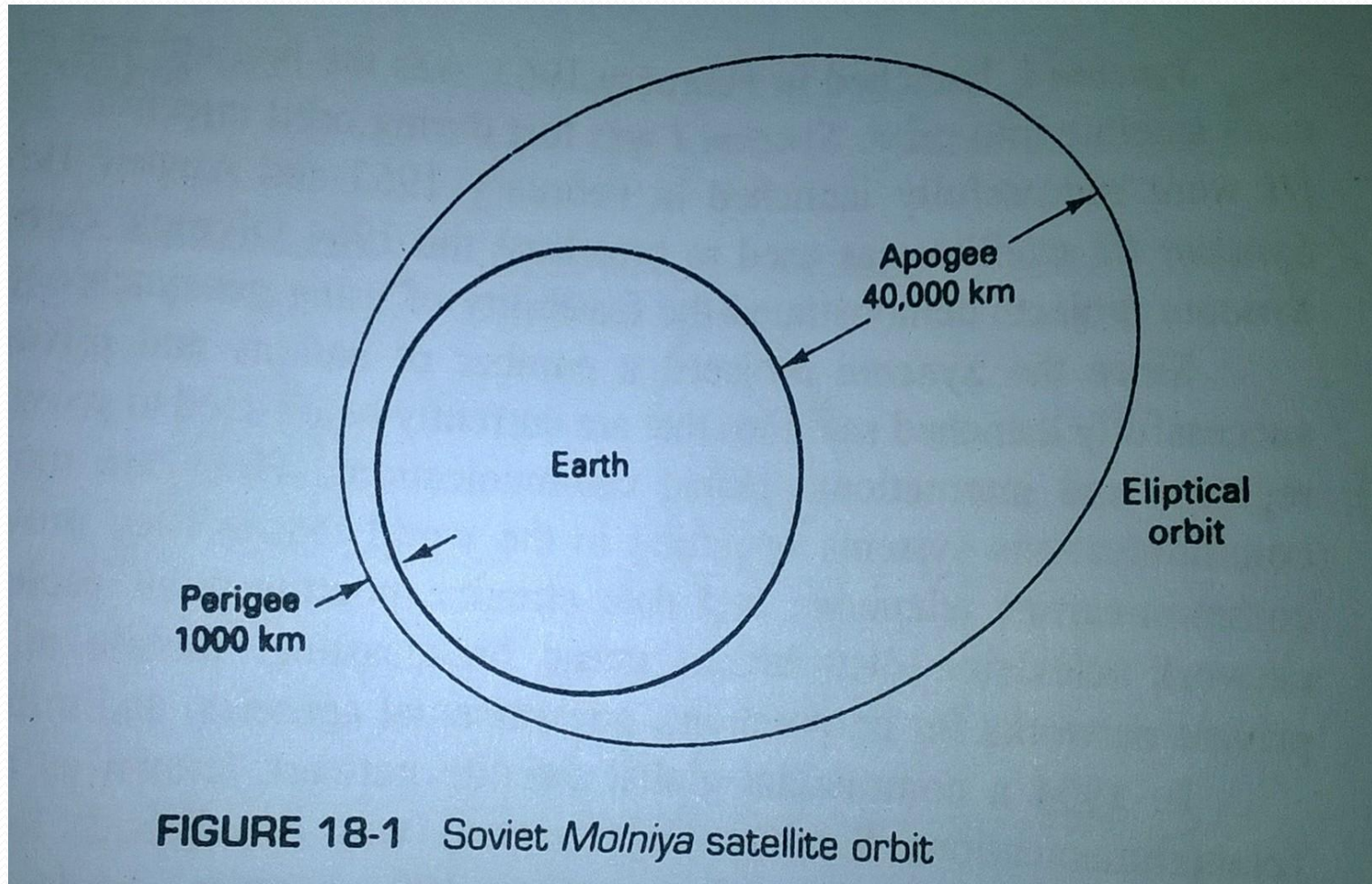
CONT...

- **Active satellite**
- An active satellite is one that electronically repeats a signal back to Earth (receive, amplifies and retransmits the signal).
- It is one in which the signal is amplified and its frequency changed by transponder, before it sent back to earth.
- Example: communication, broadcasting satellites.

Orbital satellites

- The satellites mentioned – thus far are called orbital or nonsynchronous satellites.
- **Nonsynchronous** satellites rotate around Earth in a low-altitude elliptical or circular pattern.
- One of the more interesting orbital satellite system is the **Soviet Molniya system**.
- These satellites are used for television broadcasting and are presently the only nonsynchronous-orbit commercial satellite system in use.

Soviet Molniya satellite orbit



- 
- The **apogee** is the farthest distance from Earth a satellite orbit reaches, the **perigee** is the minimum distance, and the *line of apsides* is the line joining the perigee and apogee through the center of Earth.

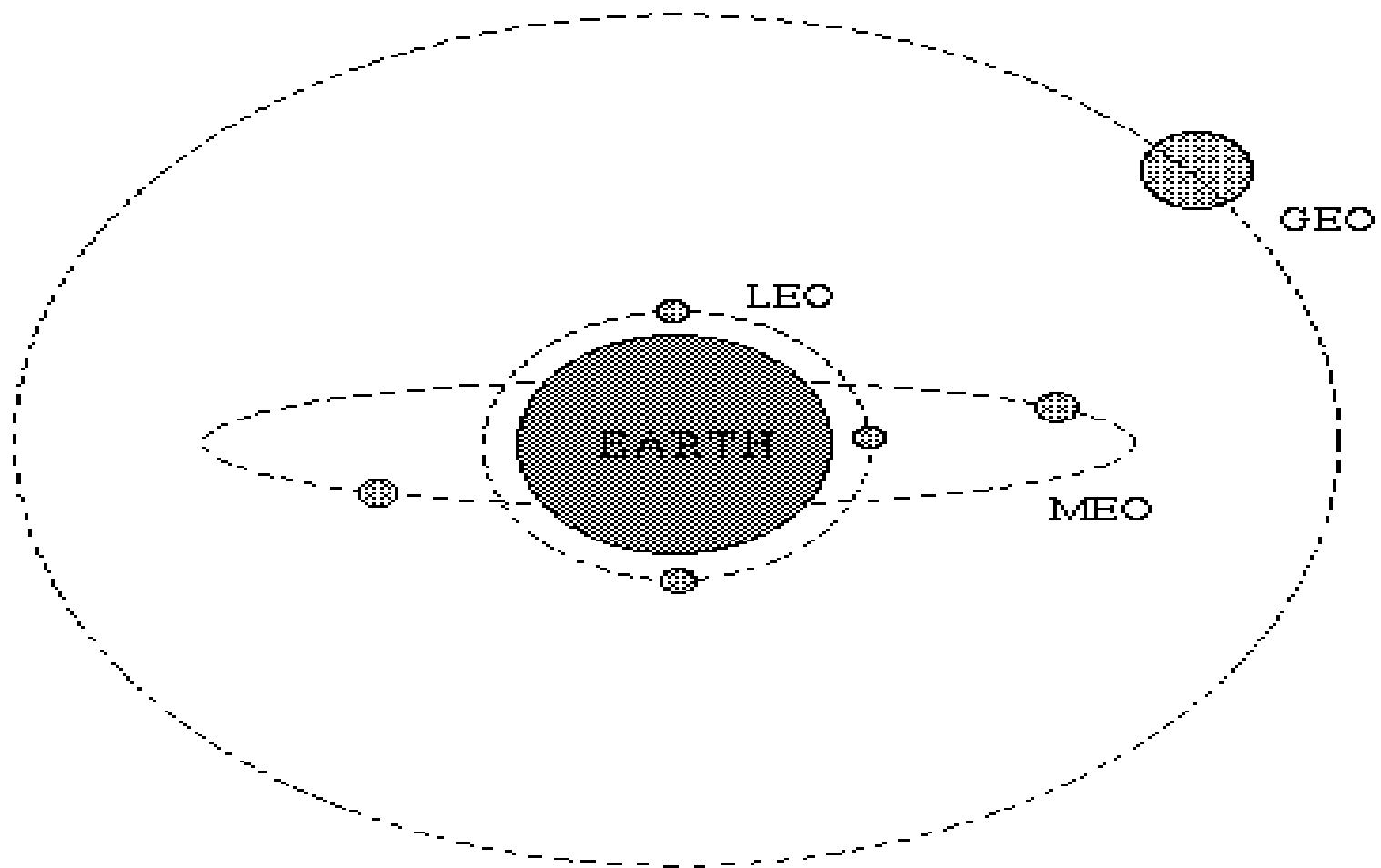
Geostationary Satellites

- Geostationary or geosynchronous satellites are satellites that orbit in a circular pattern with an angular velocity equal to that of Earth.
- Consequently, they remain in a fixed position in respect to a given point on Earth. The orbital time of a geosynchronous satellite is 24 h, the same as Earth.
- An obvious advantage is they are available to all the earth stations within their shadow 100% of the time.
- Disadvantage is they require sophisticated and heavy propulsion devices on board to keep in fixed orbit.

Orbital patterns

- Satellite orbits are also classified in terms of the orbital height. These are:
- LEO- Low Earth Orbit
- MEO- Medium Earth Orbit
- GEO- Geostationary Earth Orbit

LEO, MEO & GEO

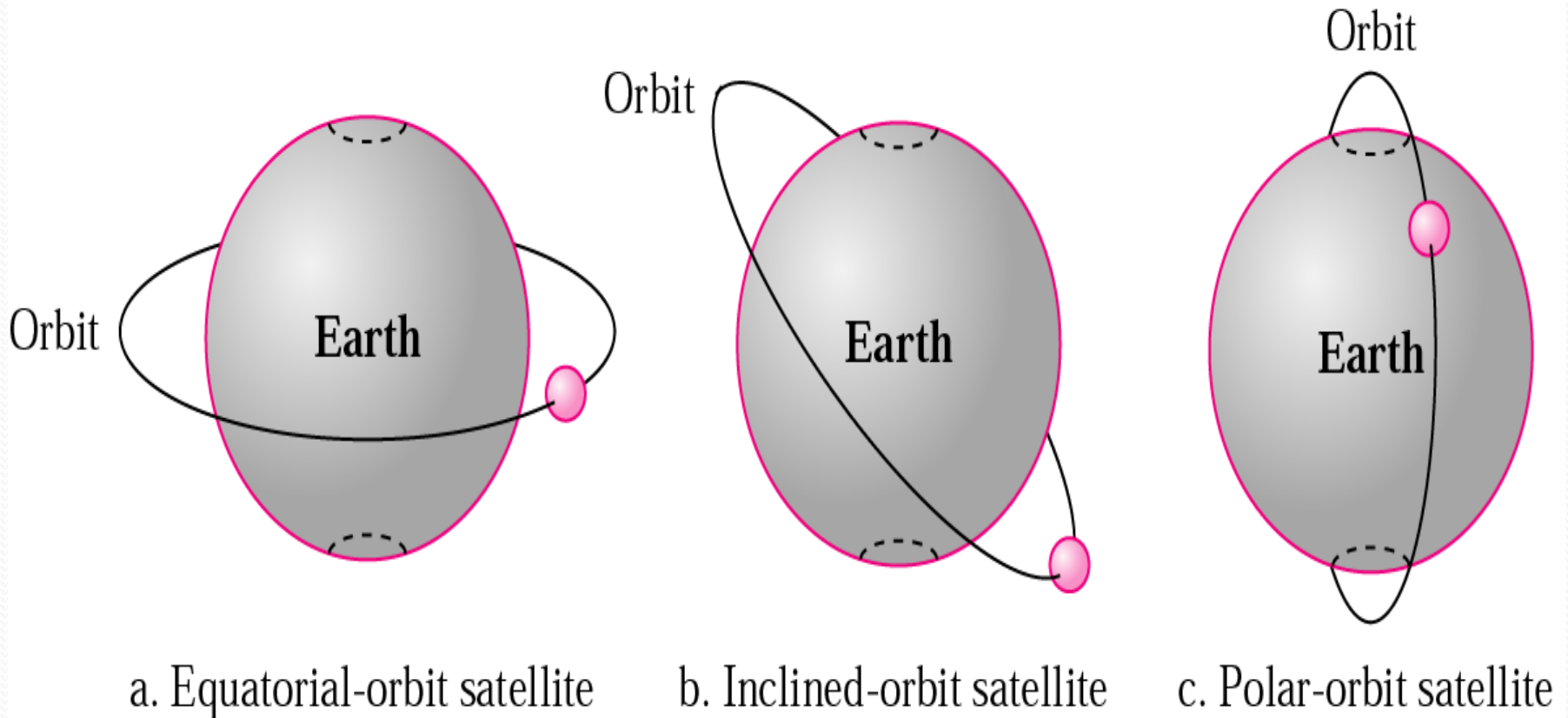


Differences

GEO	MEO	LEO
These are very far away from Earth.	These are little far from Earth.	These are much closer to Earth
These are placed at the height in between 19000 mi to 25000 mi from the Earth surface. 36000KM	These are placed at the height in between 6000 mi to 12000 mi from the Earth surface. 10,000 to 20,000 km	These are placed at the height in between 100 mi to 300 mi from the Earth surface. 160 km and 1,600 km
Revolves with same angular velocity as the Earth rotation	Revolves faster than Earth	Don't stay in fixed position relative to the surface
Maximum time rotation around 23 H: 56 M: 4S	Visible for a particular period of time, usually between 2 to 8 hours	Visible for 15 to 30 mins each pass. 1.5 hours rotate around the entire earth
Larger coverage area, a single satellite can cover 42% of earth	Not as much as Geo Satellite	Not as much as Geo and MEO satellite
Larger delay compared to other orbits	Delay is less compared to GEO orbit	Delay is lesser than GEO and MEO orbits

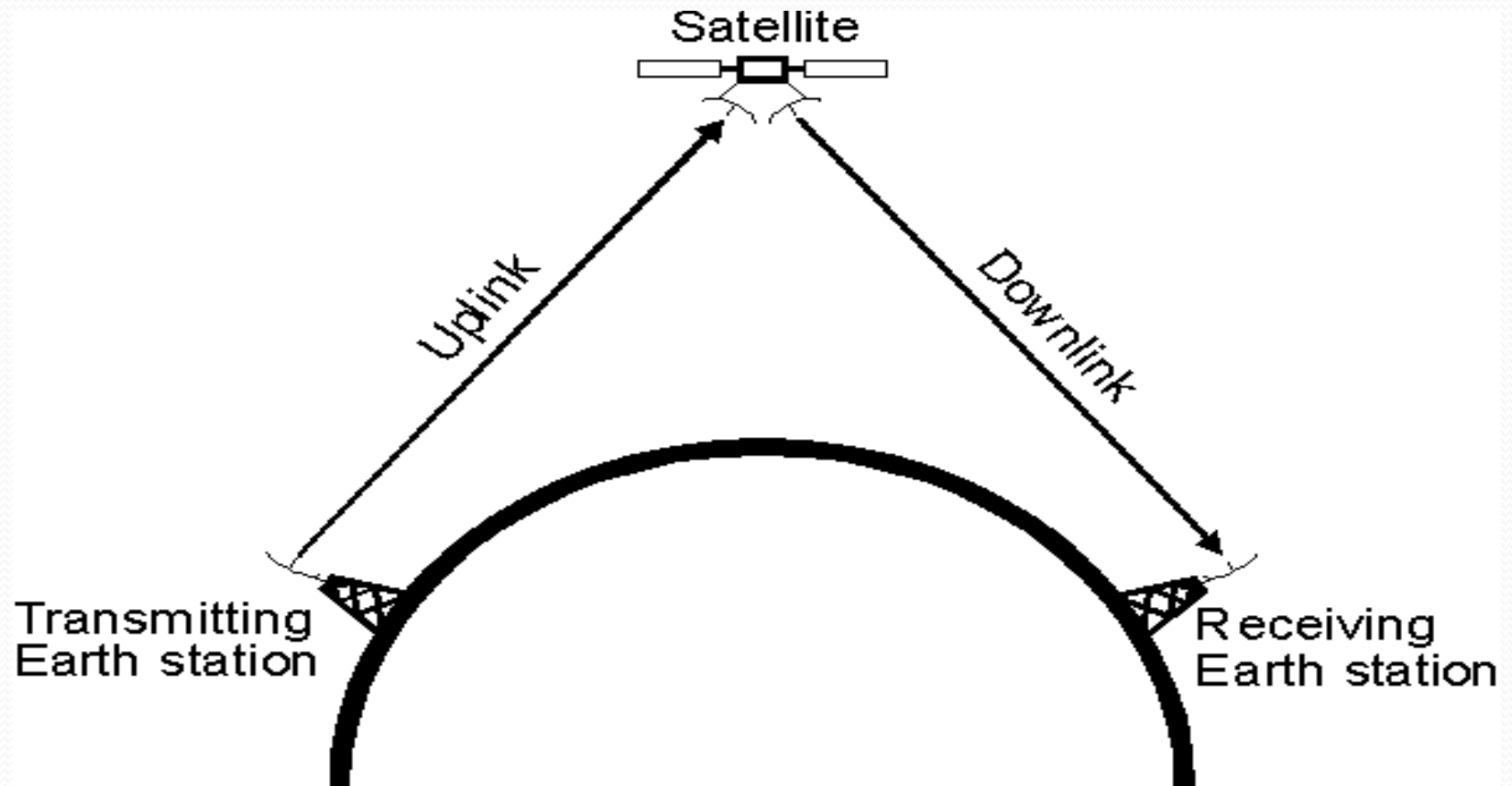
Satellite Paths

- There are three paths that a satellite may take as it rotates around the Earth.



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- **Equatorial Orbit:** When the satellite rotates in an orbit above the equator, it is called an equatorial orbit.
 - **Polar Orbit:** When the satellite rotates in an orbit that takes it over the north and south poles, it is called a polar orbit.
 - **Inclined Orbit:** Any other orbital path is called an inclined orbit.

Satellite communication picture



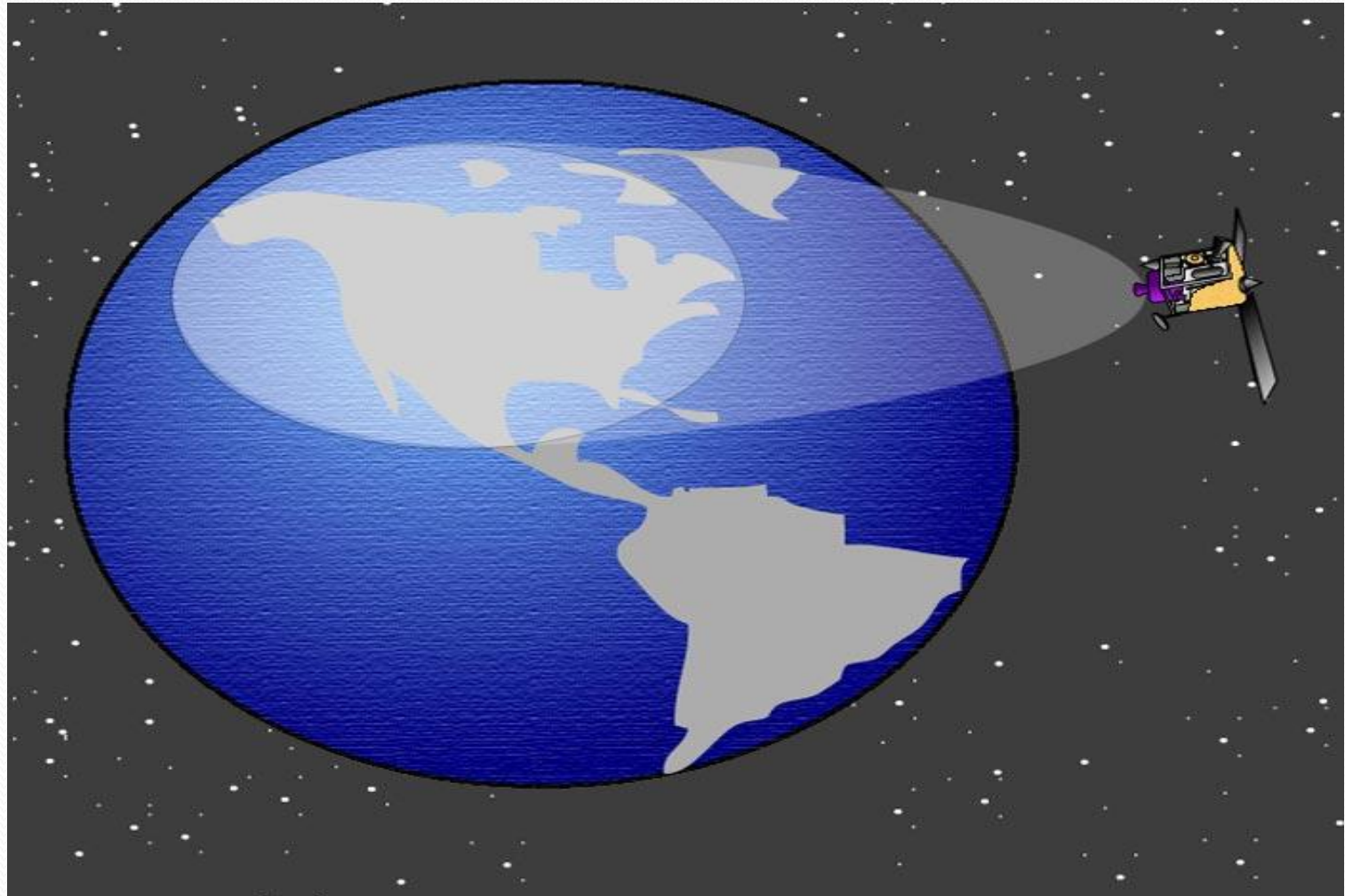
Uplinks and Downlinks

- According to the FCC, an uplink is the “transmission power that carries a signal... from its Earth station source up to a satellite; a downlink... includes the satellite itself the receiving Earth station and the signal transmitted downward between the two.”
- The uplink refers to the transmission from the Earth station to the satellite and the downlink is the transmission from the satellite to the Earth station.

Radiation Patterns: Footprints

- The area of Earth covered by a satellite depends on the location of the satellite in its geosynchronous orbit, its carrier frequency and the gain of its antennas.
- The geographical representation of a satellite antenna's radiation pattern is called a footprint.

Footprints



Footprint Categories

- The radiation pattern from a satellite antenna may be categorized as either spot, zonal or earth.
- The radiation patterns of earth coverage antennas have a beamwidth of approximately 17 degree and include coverage of approximately one-third of Earth's surface.
- Zonal coverage includes an area less than one-third of Earth's surface.
- Spot beams concentrate the radiated power in a very small geographic area.

Beams: spot, zonal or earth

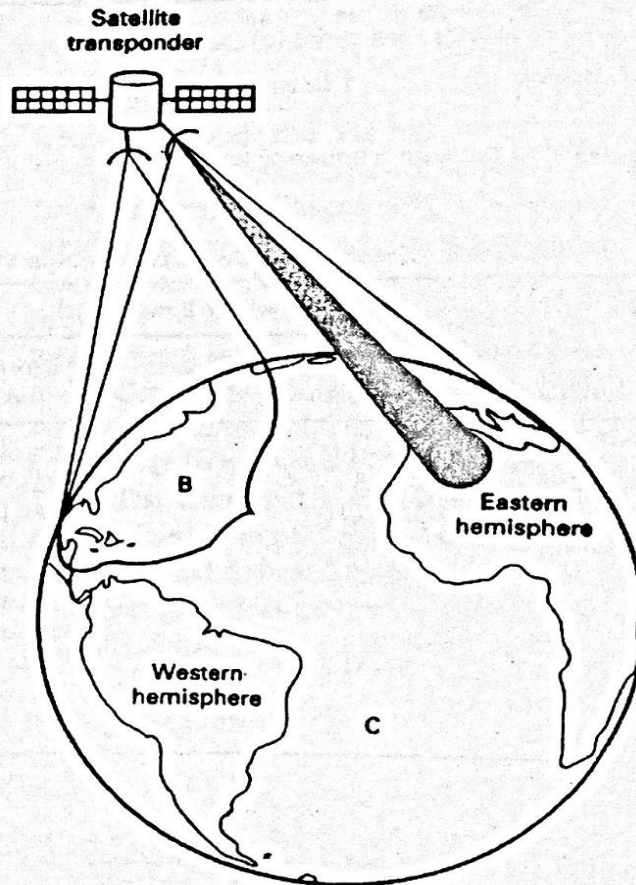


FIGURE 18-11 Beams: (a) spot; (b) zonal; (c) earth

Satellite Transmission Bands

Frequency Band	Downlink	Uplink
C	3.7-4.2 GHz	5.92-6.42 GHz
Ku	11.7-12.2 GHz	14.0-14.5 GHz
Ka	17.7-21.2 GHz	27.5-31.0 GHz

The C band is the most frequently used. The Ka and Ku bands are reserved exclusively for satellite communication but are subject to rain attenuation

Satellite Frequency Bands and Antennas (Dishes)



C-Band

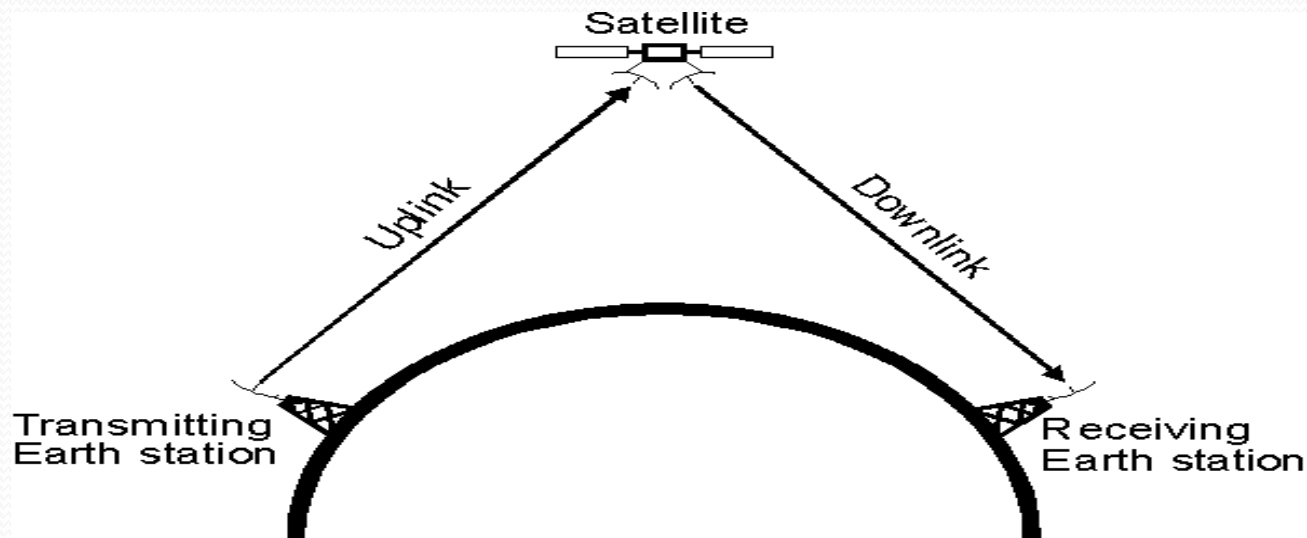


Ku-Band

- Most commonly used bands: C-band (4 to 8 GHz) , Ku-band (11 to 17 GHz) , and Ka-band (20 to 30 GHz).

Satellite System

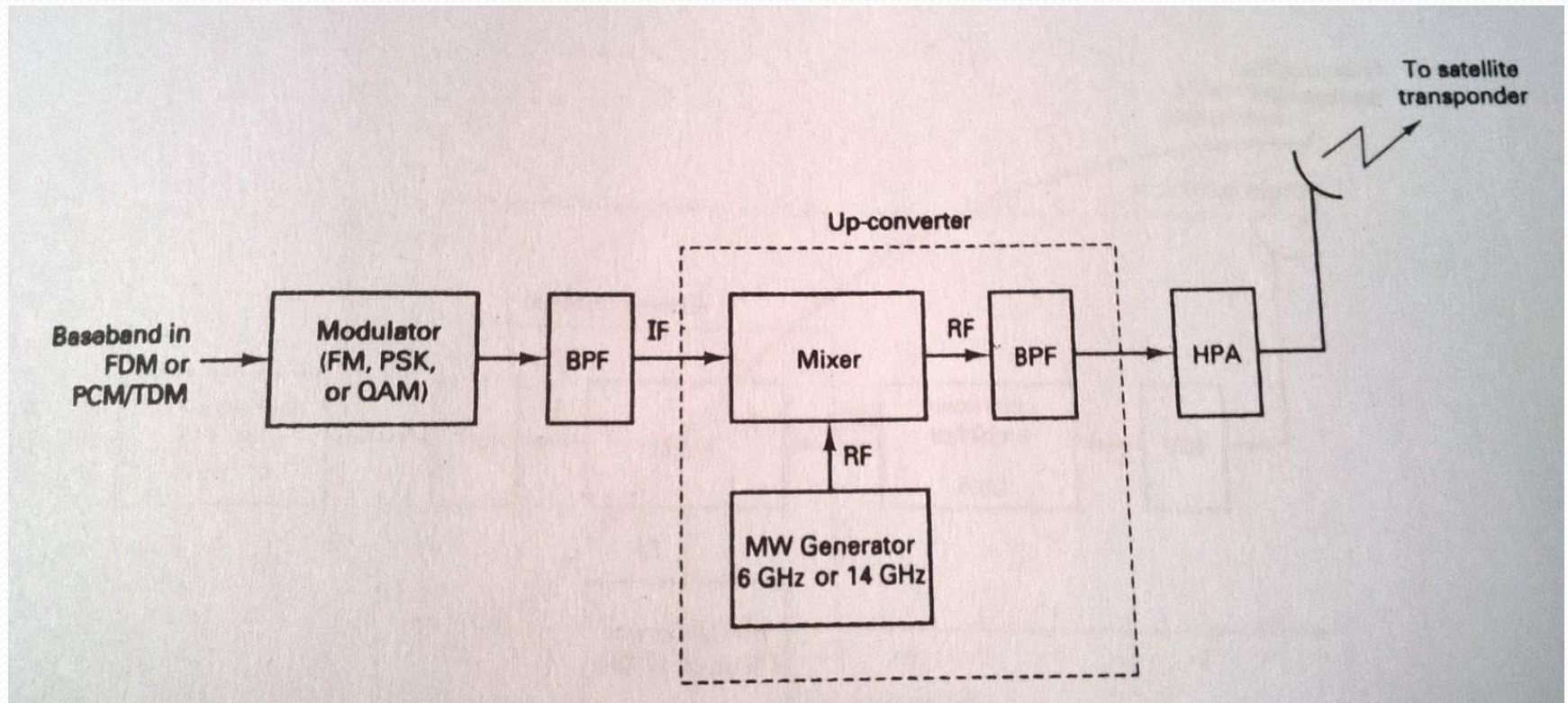
- Essentially, a satellite system consists of 3 sections:
- **An up-link**
- A satellite transponder
- And a down-link



Up-link Model

- The primary component within the up-link section of a satellite system is the earth station transmitter.
- A typical earth station transmitter consists of an IF modulator, an IF-to-RF microwave up-converter, a high-power amplifier (HPA), and some means of band limiting the final output spectrum.

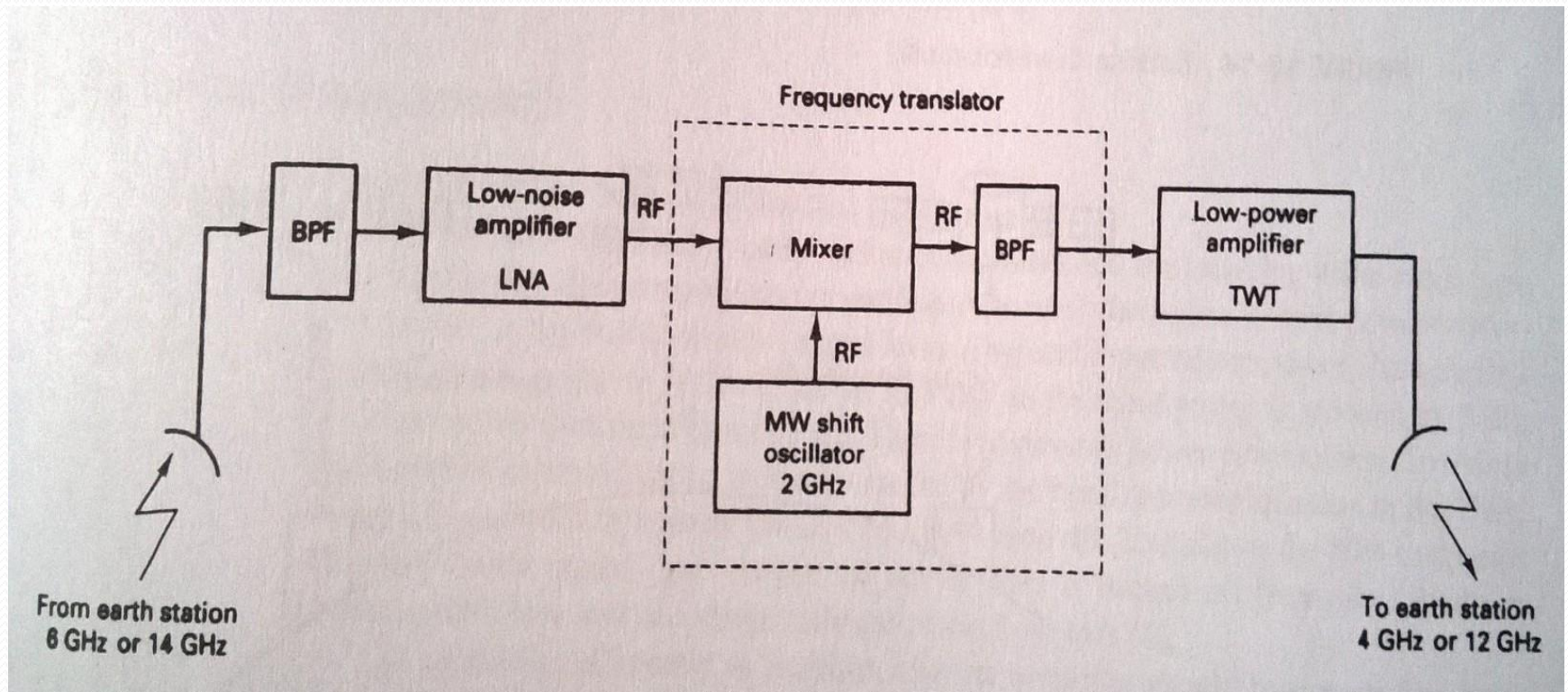
Satellite up-link model



Transponder

- A typical satellite transponder consists of an **input bandlimiting device (BPF)**, an **input low-noise amplifier (LNA)**, a **frequency translator**, a **low level power amplifier**, and an **output bandpass filter**.

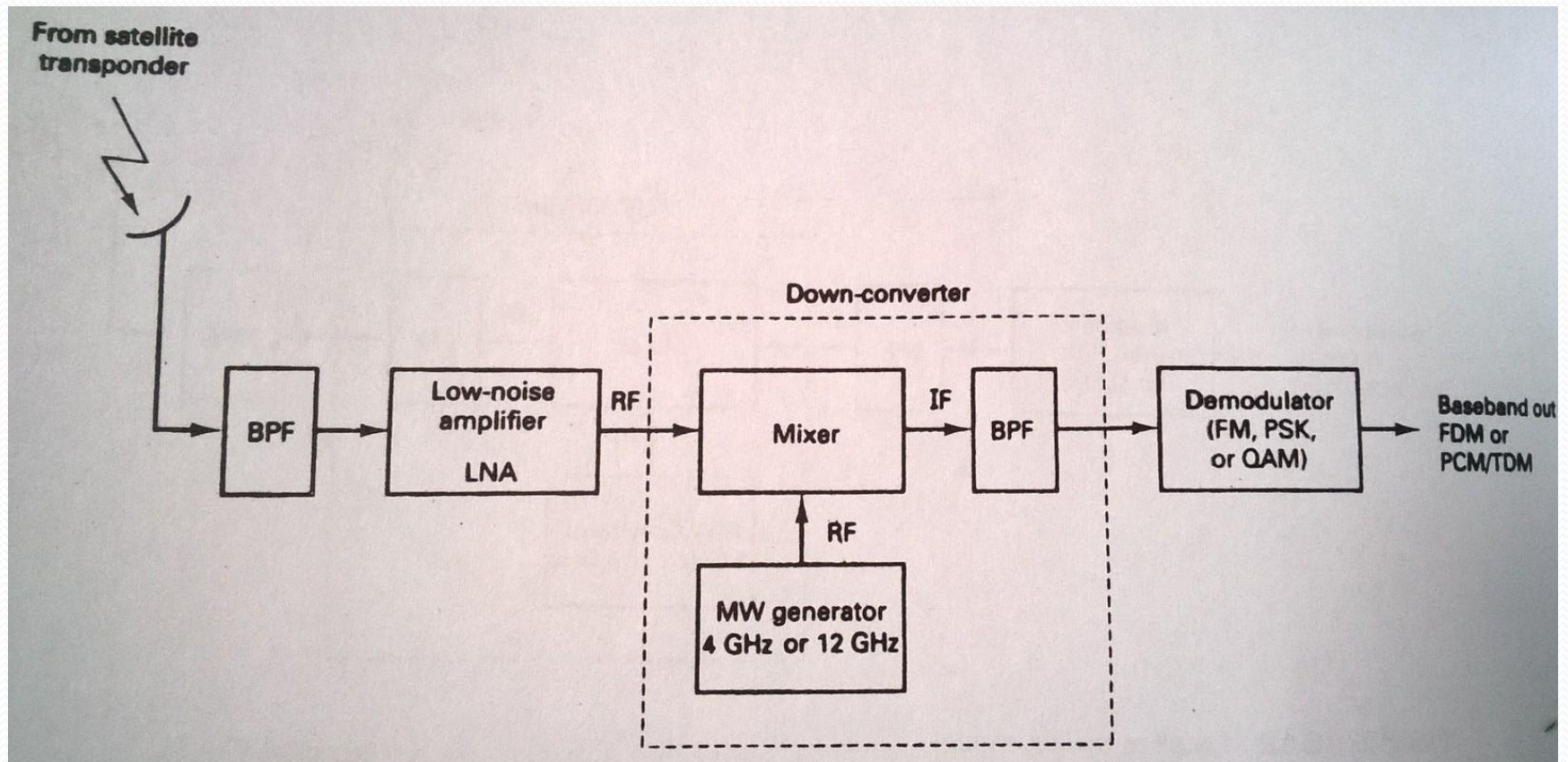
Satellite Transponder



Down-link Model

- An earth station receiver includes an **input BPF**, an **LNA**, and an **RF-to-IF down-converter**.

Satellite down-link model



Cross-links

- Occasionally, there is an application where it is necessary to communicate between satellites. This is done using satellite cross-links or intersatellite links (ISLs).

Cross-links

